U.PORTO
FEUP FACULDADE DE ENGENHARIA
UNIVERSIDADE DO PORTO

MESTRADO EM CIÊNCIA DA INFORMAÇÃO

The Large Language Models and Ethical Issues in Scientific Research

Renata Filipa Moreira Mendes

M
2025

PREVIEW

UNIDADES ORGÂNICAS ENVOLVIDAS

FACULDADE DE ENGENHARIA
FACULDADE DE LETRAS

FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

FACULDADE DE LETRAS DA UNIVERSIDADE DO PORTO

The Large Language Models and Ethical Issues in Scientific Research

Renata Filipa Moreira Mendes

PREVIEW

Para avaliação por Júri

Mestrado em Ciência da Informação

Supervisor: Professor Nuno Feixa Rodrigues

June, 2025

PREVIEW

© Renata Filipa Moreira Mendes, 2025

The Large Language Models and Ethical Issues in Scientific Research

Renata Filipa Moreira Mendes

Mestrado em Ciência da Informação

PREVIEW

Approved by the Doctoral Committee:

President: Dr. Gabriel de Sousa Torcato David, Associate Professor

Referee: Dr. Fábio Mendonça, Assistant Professor

Supervisor: Dr. Nuno Feixa Rodrigues, Coordinator Professor

June 27, 2025

Abstract

In the current information age, rapid advances in Artificial Intelligence (AI) and Machine Learning (ML) are transforming countless disciplines. This dissertation explores the emerging role of Large Language Models, such as GPT-4 and AutoGPT, as innovative tools that can generate human-like text and enhance scientific research processes. Positioned at the intersection of computational efficiency and intellectual enquiry, LLMs offer unparalleled opportunities to automate labour-intensive tasks traditionally associated with scientific research. While these models have the potential to reshape conventional research paradigms and adapt rapidly to the evolving demands of the discipline, their use introduces complex ethical dilemmas. These range from algorithmic bias and misinformation to concerns about data privacy and transparency, and the erosion of accountability in scientific results.

This dissertation aims to critically evaluate the capabilities and limitations in scientific research and explore their potential for responsible integration into research workflows.
The study includes a scoping review, conducted following PRISMA guidelines. Multiple online databases, such as SCOPUS, ERIC, EBSCO, and Web of Science (WoS), were searched for relevant literature, focusing on studies that utilize these models in scientific contexts.

A survey was conducted among researchers from INESC-TEC and Master's students involved in research activities utilizing LLMs, to identify opinions of the current applications, ethical challenges, and best practices in research. The survey collected qualitative and quantitative data on research aims, findings, and ethical considerations related to their use.

While Large Language Models hold significant potential to advance scientific research, their responsible integration necessitates careful ethical considerations. This study highlights the positioning of researchers in earlier and more advanced areas of scientific research in relation to the use in their research activities.

Keywords: LLMs, artificial intelligence, ethical guidelines, scientific research

PREVIEW

UN Sustainable Development Goals

Completing a Master's dissertation in Science Information represents not only a significant academic milestone but also an opportunity to contribute meaningfully to the global agenda for sustainable development. By analyzing my project through the lens of United Nations SDGs, I recognize its potential to addresses SDG 16, which promotes peaceful, just, and inclusive societies, by exploring the ethical challenges related to the use of advanced technologies in science.

Specifically, my work contributes to the following targets : promoting the responsible use of artificial intelligence and strenghthening ethical frameworks in research. By investigating ethical issues such as bias, privacy, and transparency, my project aims to ensure that advances in language models benefit society equitably, promoting digital inclusion and social justice.

The positive impacts include fostering responsible practices in technology development, increasing awareness of ethical risks, and assist in the formulation of guidelines that guide the ethical use of AI in scientific research.

Reflecting on my research from the SDG perspective, I recognize that my work can act as a catalyst for a more ethical, transparent, and inclusive science, contributing to a future where technology is a tool for positive social transformation and sustainable development.

PREVIEW

SDG	Goal	Contribution	Performance indicators and metrics
16	16.6	Promote transparency and accountability in the use of LLMs in scientific research	Scientific publications that disclose the use of LLMs and detail their application and through the survey, where it reports the opinions of researchers and their contribution to the construction of ethical guidelines

Acknowledgements

A realização desta dissertação de mestrado representa uma etapa fundamental na minha trajetória académica e pessoal e não poderia deixar de expressar a minha profunda gratidão a todos aqueles que, de forma direta e indireta, contribuíram para que este sonho se tornasse possível. Sem o apoio, incentivo e orientação de cada um, este percurso não teria sido tão enriquecedor nem tão realizador.

Em primeiro lugar, gostaria de agradecer aos professores Nuno Feixa Rodrigues e António Coelho, pela orientação principalmente em momentos de maior desorientação, pela disponibilidade fornecida ao longo deste trajeto, pelo conhecimento partilhado e pelas críticas construtivas que ampliaram os meus horizontes para a criação e elaboração deste trabalho. A vossa dedicação não passou despercebida e deixou uma marca indelével na minha formação académica.

Aos meus amigos, que sempre estiveram ao meu lado em todas as fases da minha vida, incluindo esta, deixo o meu mais sincero agradecimento. À Maria, ao Tiago, ao Gervásio e ao Bruno, por todas as conversas e pelo apoio contínuo. Sou muito grata por poder partilhar esta conquista convosco.

À minha família, dedico um agradecimento especial e profundo. À minha mãe, que é o meu amor alicerce, pela dedicação, amor incondicional e por tudo o que faz para me ver feliz. A tua força, coragem e exemplo de perseverança continuam a ser a minha maior inspiração. Ao meu irmão Daniel, por todo o carinho, pelas palavras de incentivo e pelo exemplo de dedicação ao trabalho árduo. Ao meu avô, por acreditar em mim, por me motivar a persistir e a nunca desistir dos meus sonhos. E por fim, ao Nicolau, agradeço do fundo do coração, por todo o carinho, paciência e apoio dado em todas as fases deste projeto e por sobretudo, ter acreditado em mim e nunca me ter deixado desistir.

A todos vocês, obrigada por acreditarem em mim, por compreenderem as minhas ausências e por fazerem parte desta minha conquista. Estou eternamente grata.

Renata Moreira

PREVIEW

PREVIEW

"One day, in retrospect, the years of struggle will strike you as the most beautiful."

Sigmund Freud

Contents

Introduction **1**
1.1 Context and Motivation 2
1.2 Goals 3
1.3 Methodology 4
1.4 Data collection and analysis techniques 4
1.5 Document Structure 5

Literature review **7**
2.1 Research Methodology 7
2.2 PRISMA Diagram 9
2.2.1 PRISMA results 10
2.2.2 PRISMA results 11
2.2.3 PRISMA results 12
2.3 Data Collection 14
2.3.1 Discussion 17
2.4 Summary 19

Ethical Management of Large Language Models in the Scientific Research Process21
3.1 Large Language Models 22
3.1.1 Capabilities and Limitations of LLMs 26
3.1.2 Applications of LLMs in Scientific Research 28
3.2 Ethical Issues in Scientific Research related to Large Language Models 35
3.3 Ethical Guidelines 37
3.4 Summary and Conclusions 40

Problem Statement **43**

Survey **45**
5.1 Form Section 1 – Participants Characteristics 45
5.2 Form Section 2 – Questionnaire with Likert Scale 47
5.3 Form Section 3 – Participants Open Answers 49
5.4 Design of the Survey 50

Data Analysis of the Survey **52**
6.1 Analysis Section 1 – Participants Characteristics 52
6.2 Analysis Section 2 – Questionnaire with Likert Scale 54

PREVIEW

6.3 Analysis Section 3 – Participants Open Answers 75
6.4 Discussion 79

Conclusions 81
7.1 Research questions 81
7.2 Future Work 83

Bibliography 84

Attachments 89

PREVIEW

List of Figures

Figure 1- PRISMA Diagram Haddaway et al., 2022 9
Figure 2 - Percentage of the level of qualification in scientific research 53
Figure 3 - Percentage of the time of scientific research activities 53
Figure 4 - Percentage of the number of scientific articles published 54
Figure 5 - Question 1: Likert Scale Results 55
Figure 6 - Question 2: Likert Scale Results 56
Figure 7 - Question 3: Likert Scale Results 57
Figure 8 - Question 4: Likert Scale Results 58
Figure 9 - Question 5: Likert Scale Results 59
Figure 10 - Question 6: Likert Scale Results 60
Figure 11 - Question 7: Likert Scale Results 61
Figure 12 - Question 8: Likert Scale Results 62
Figure 13 - Question 9: Likert Scale Results 63
Figure 14 - Question 10: Likert Scale Results 64
Figure 15 - Question 11: Likert Scale Results 65
Figure 16 - Question 12: Likert Scale Results 66
Figure 17 - Question 13: Likert Scale Results 67
Figure 18 - Question 14: Likert Scale Results 68
Figure 19 - Question 15: Likert Scale Results 69
Figure 20 - Question 16: Likert Scale Results 70
Figure 21 - Question 17: Likert Scale Results 71
Figure 22 - Question 18: Likert Scale Results 72
Figure 23 - Question 19: Likert Scale Results 73
Figure 24 - Question 20: Likert Scale Results 74

PREVIEW

List of Tables

Table 1- Keywords 8
Table 2 : Number of results per query 10
Table 3 : Number of queries per database 10
Table 4 : Exclusion Criteria 11
Table 5 : Number of results per query 11
Table 6 : Number of queries per database 12
Table 7 : Inclusion Criteria 13
Table 8: Number of results per query 13
Table 9 : Number of queries per database 13
Table 10 - The evolution of LLMs Waduge, A. O. O. et.al., 2024 25
Table 11 - List of AI tools for researchers Lin, Z. C., 2023 33

PREVIEW

Abreviaturas e Símbolos / Abbreviations and symbols

LLMS	Large Language Models
AI	Artificial Intelligence
ML	Machine Learning
NPL	Natural Language Processing

PREVIEW

Chapter 1

Introduction

Artificial Intelligence (AI) and Machine Learning (ML) have emerged as transformative forces in recent decades, reshaping various sectors, including commerce, health and education. One of the most impactful developments of this technological wave is the emergence of Large Language Models, such as GPT-4 and AutoGPT, which can generate sophisticated, human-like text. These models have aroused considerable interest, not only because of their technical capabilities, but also because of their potential to revolutionize traditional scientific research processes – especially in fields that rely heavily on information processing such as engineering.

While AI has been widely discussed in the context of automation and decision-making, the application of these models specifically as tools in academic and scientific contexts remains a developing area of research. Their use is both promising and presents considerable ethical uncertainty. As these models begin to take on functions traditionally reserved for human researchers – such as writing literature reviews, analysing data trends and even creating hypotheses – it becomes imperative to examine the consequences of their use. At the heart of this discussion involves understanding not only the practical advantages, but also the ethical challenges they introduce, including bias, lack of transparency and issues related to authorship and responsibility.

This leads to the central research question of this study: "How are researchers currently utilizing Large Language Models in scientific research?".

Despite LLMs being increasingly implemented in various research environments, a consensus has yet to be reached on how to use them correctly, what their limitations are, and what frameworks are needed to ensure their ethical integration. This highlights a timely research opportunity, particularly in scientific research, where methods must continually evolve to keep pace with technological developments. Given the invitation to explore this theme through an academic lens, this dissertation aims to critically evaluate the role of these models in scientific research

PREVIEW

– focusing not only on their contributions to efficiency and innovation, but also on the broader implications for research integrity and ethical standards.

This scoping review seeks to systematically map the existing literature on the use of LLMs in scientific research, identify key themes, and highlight gaps in current knowledge. By doing so, it aims to provide a comprehensive overview of how these models are being integrated into research practices, the ethical and methodological challenges faced, and the potential future directions for their responsible adoption.

In addition to this central discussion, the dissertation will reflect on the survey that concludes the study, exploring how their presence can impact the lives of the scientific community, from master´s students to senior scientific researchers, raising opinions on the advantages and disadvantages of using Large Language Models and some critical considerations on authorship and the need for ethical guidelines.

1.1 Context and Motivation

This theme was proposed by INESC-TEC, currently recognised as a private, non-profit association with public utility status. It focuses on scientific research and technological development activities, technology transfer, advanced consultancy and training, and the pre-incubation of new, technology-based companies. Its mission is to promote scientific and technological advancement, as well as innovation based on technology, by transferring new knowledge and technologies to industry, services and public administration.

In line with its strategic focus on digital transformation and intelligent systems, INESC-TEC has identified the growing impact of artificial intelligence, particularly Large Language Models as a key area of interest and concern. These models are beginning to permeate research workflows, raising important questions about how they should be integrated into scientific settings. While LLMs can significantly enhance productivity and support research activities, the organisation recognises that their use also introduces ethical challenges that must be addressed to preserve scientific integrity.

The present dissertation, carried out within the scope of the Master's in Information Science, integrated into the Faculty of Engineering and the Faculty of Arts of the University of Porto, was developed with a focus on evaluating the impact of these tools on academic research, exploring their potential as new tools to cultivate research methodologies.

The transition from a Bachelor's degree in Information Science to a Master's degree in the same field represents a progression and continuous improvement in the pursuit of more knowledge about the vast area of information. My motivation lies in consolidating all the knowledge and skills acquired between the bachelor's and master's degrees to carry out this work.

PREVIEW

When it came time to choose the topic and area of study for the dissertation, the interest in the area proposed by INESC-TEC stood out, as it addresses certain challenges that occur in our daily lives, which need to be explored, investigated, and discussed. The fact that it is a relevant issue today and involves the information age was also an important factor in choosing this project.

This research was motivated by concerns about the increasing integration of large language models in scientific research environments. As these advanced tools become more common, their potential to revolutionize research by enhancing productivity, facilitating data analysis, and supporting hypothesis generation becomes clearer. However, there are significant issues that need careful consideration. Their development in academic settings raises important questions about ethics, transparency, and research integrity, which must be addressed to ensure responsible use.

The motivation for this work includes recognizing that, while these models can assist in various research phases such as literature review, data interpretation, and drafting scientific texts there are risks of unintended consequences. These include embedded biases, potential misuse or overreliance on automated outputs, and the opaque nature of the algorithms. Such problems can undermine the credibility of findings and threaten ethical standards essential for rigorous research.

In the Problem Statement section, the study will explore these challenges more deeply by outlining specific issues associated with integrating these models into research workflows. The analysis will be based on a review of current developments in AI technology, the benefits of their use, and existing guidelines for ethical application.

PREVIEW

1.2 Goals

The primary goal of this research is to explore how researchers currently utilize large language models in scientific environments. Specifically, the study aims to answer: “How are researchers using these models in their work?” This investigation seeks to map existing applications, practices, and perceptions related to their integration across various research disciplines.

Building on this, the broader objective is to analyze how these tools can be effectively and ethically incorporated into research workflows. This includes assessing their potential to automate or improve tasks such as hypothesis generation, literature review synthesis, content creation, and data analysis. The study aims to highlight both the benefits and efficiencies gained, as well as critically examine the ethical considerations involved.

While the main focus is on providing an overview of current practices and challenges, this work also aims to lay the foundation for future research into more detailed aspects of their integration. It recognizes the transformative potential of these models for research methodologies and seeks to contribute to the ongoing discussion about their responsible use in academic settings.

Ultimately, this work intends to serve as a reference for the scientific community, offering insights that support informed decision-making and responsible implementation ensuring that innovation advances alongside ethical standards.

1.3 Methodology

Given the nature and objectives of this dissertation, the chosen methodological approach is based on the interpretive paradigm, which offers a more human-centered and qualitative perspective. This paradigm is particularly well-suited for studies examining social, ethical, and organizational dimensions—such as the implications of integrating Large Language Models into scientific research practices. The interpretive approach promotes reflection, interaction, and critical analysis, allowing the researcher to engage deeply with both theoretical concepts and practical contexts.

This dissertation addresses not only the capabilities of LLMs but also explores the behaviors, perceptions, and ethical concerns of individuals and institutions involved in research processes. Understanding how researchers perceive and utilize these tools, as well as how institutions approach governance of AI technologies, is central to identifying risks and proposing responsible, viable solutions.

To gain a comprehensive understanding of the current landscape, the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocols) platform was employed to conduct a scoping review. This initial phase helped identify key concepts, challenges, and ethical issues related to the use of LLMs in scientific research. Following this, the study included contextual analysis and informal interviews with individuals directly engaged in relevant processes, providing valuable insights into practical realities and perceptions.

Considering this approach, the survey methodology specifically, using questionnaires—was deemed the most appropriate. As a branch of applied statistics focused on human research surveys, survey methodology involves selecting samples from a population and employing techniques for collecting survey data, including questionnaire design and methods to improve response rates and accuracy (Groves, R. M., et al., 2004). After evaluating various research approaches, the survey method was selected to gather comprehensive insights into the perceptions, concerns, and attitudes of researchers at different levels.

1.4 Data collection and analysis techniques

In the initial phase of this research, a comprehensive literature review was conducted to gather foundational knowledge on Large Language Models and their associated ethical

PREVIEW

considerations within the context of scientific research. This review was scoping carried out using PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology, which ensures a transparent and rigorous process for identifying, screening, and selecting relevant studies.

The PRISMA process involved several steps: first, a broad search performed across multiple academic databases using defined keywords related to LLMs and ethics. Subsequently, duplicates were removed, and titles and abstracts were screened to exclude irrelevant studies. The remaining articles underwent full-text review to determine their relevance and quality. This scoping approach facilitated the synthesis of current knowledge, identifying key themes, gaps, and ethical challenges in the fields.

Building upon the state-of-art review, the research progressed to the development and dissemination of a survey. The survey was designed to gather insights from researchers, regarding their perceptions in the use of these models in scientific research. The data collected through the survey was then analysed quantitatively and qualitatively to identify opinions and potential areas for investigation.

1.5 Document Structure

In addition to the introductory topics, this dissertation is organized into the following chapters: Chapter 2 – Research Methodology, which outlines the methodological approach adopted for the literature search and analysis; Chapter 3 – Literature Review and State of the Art, where the current knowledge and key concepts related to Large Language Models, research ethics, and relevant frameworks are examined; Chapter 4 – Problem Statement, which provides a detailed explanation and contextualization of the central problem addressed in this study; Chapter 5 – where the design of the survey is presented; Chapter 6 – Data analysis of the survey and Chapter 7 – Conclusion, which summarizes the information retrieved and future directions of the research. The dissertation concludes with the complete list of references that supported this work.

PREVIEW

Chapter 2

Literature review

To establish a systematic and rigorous approach to the literature review, the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) platform was utilized. This methodology provided a structured framework to ensure transparency, reproducibility, and understanding of the review process. By following PRISMA guidelines, it was possible to identify, select, and critically appraise relevant scholarly works in a systematic manner, enabling a thorough exploration of existing research related to LLMs and ethical issues in scientific research.

PREVIEW

2.1 Research Methodology

For the bibliographic review, some academic databases were selected based on their relevance, coverage, and credibility in scientific and technological research. The primary databases included SCOPUS, ERIC, EBSCO, and Web Of Science (WoS). These platforms were appropriate due to their extensive collections of peer-reviewed articles and scholarly publications that are pertinent to the research domain.

Following the selection of databases, specific keywords and search terms were developed to formulate precise queries. These keywords reflected the core concepts of the study, namely large language models, ethical considerations, and scientific research practices. The keywords were organized into a table for clarity:

A	B	C
Large Language Models	Ethical issues	Scientific Research
LLMS	Ethical guidelines	
Artificial Inteligence	Ethics	

Table 1- Keywords

Using these keywords, a series of search queries were constructed to optimize data retrieval from each database. The goals was to maximize relevance and comprehension, ensuring that the most pertinent literature was captured. The specific search strings employed across databases are as follows:

1. “large language models” AND “ethical issues” AND “scientific research”
2. “large language models” AND “scientific research” AND “ethics”
3. (“lllms" OR "artificial intelligence”) AND "scientific research" AND "ethical issues"
4. "artificial intelligence" AND "ethics" AND "scientific research"
5. "artificial intelligence" AND "scientific research" AND "ethical guidelines"

All queries were executed across each selected database, although the volume of results varied significantly depending on the platform. To ensure the relevance and currency of the literature, a temporal filter was applied, restricting the search to publications from the last five years (2020-2025). This frame was chosen to capture the most recent developments, trends, and debates pertaining to the intersection of LLMs, ethics, and scientific research, providing a contemporary foundation for the review.

The PRISMA diagram is presented below:

PREVIEW

2.2 PRISMA Diagram

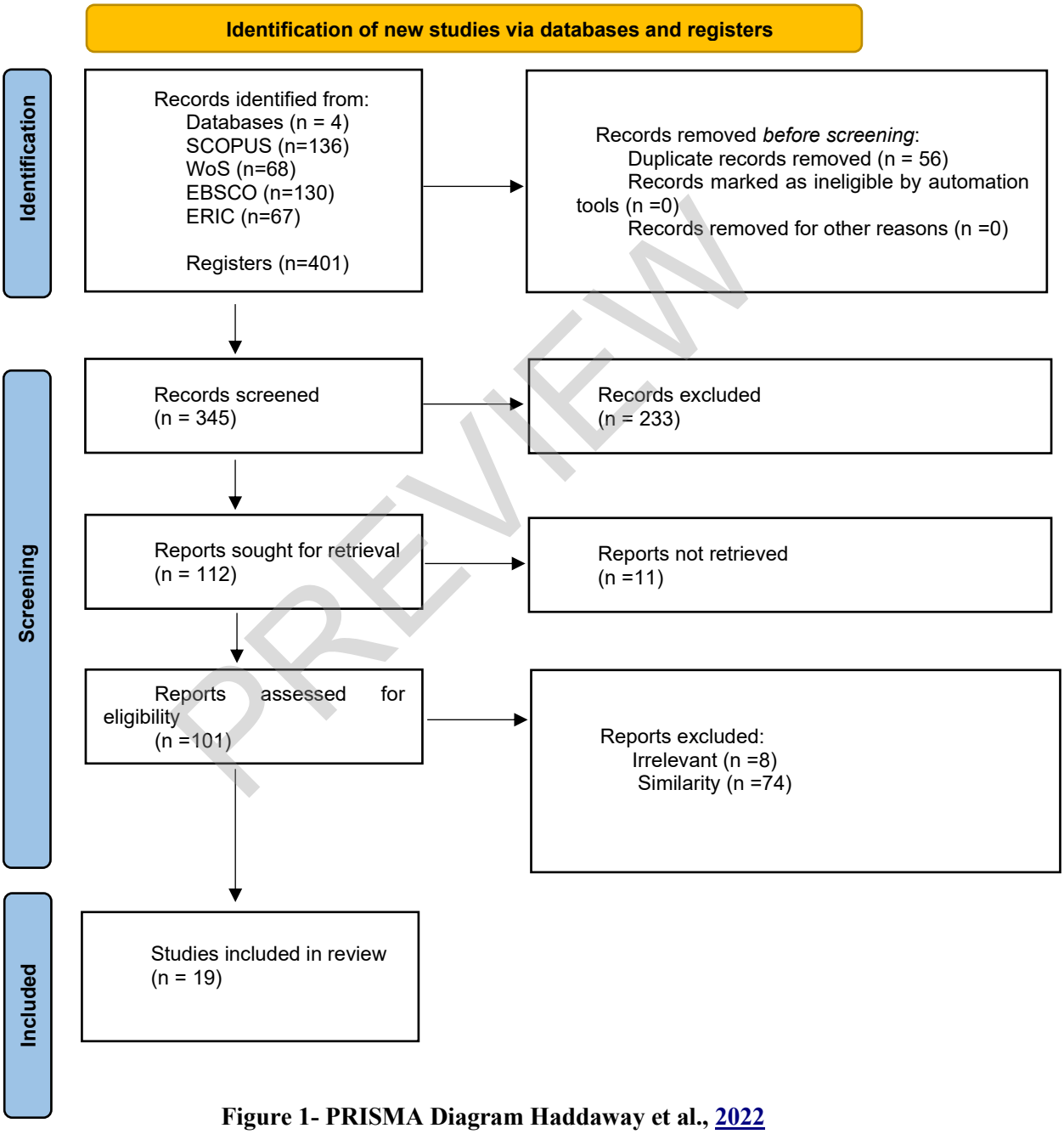

Figure 1- PRISMA Diagram Haddaway et al., 2022

PREVIEW

2.2.1 PRISMA results

According to the previously established criteria, after retrieving all results from the databases, 401 results were retrieved and added to the EndNote platform. To organize the collected data, groups were created within the platform, named “Q1,” “Q2,” “Q3,” “Q4,” and “Q5” (previously created queries) and subgroups with the names of the databases used (“SCOPUS,” “ERIC,” “EBSCO,” and “WoS”).

Once the data was correctly located in each group and subgroup, we chose to use an EndNote feature that automatically eliminates duplicate documents, removing a total of 56 documents.

Google Scholar is not included in these phases because, due to the large number of results for each query, those relevant to the topic in question were selected directly.

In this first stage, the results for each query were as follows:

Q1	Q2	Q3	Q4	Q5
17	68	92	195	29

Table 2 : Number of results per query

Once the number of results per query has been identified, it is also important to reveal the number of queries retrieved by each database:

SCOPUS	WoS	EBSCO	ERIC
136	68	130	67

Table 3 : Number of queries per database

In order to compare the results of the data collection from each database, we can see that the highest percentage is found in the SCOPUS database and the one with the fewest results is the ERIC database.

PREVIEW

2.2.2 PRISMA results

In this second stage of PRISMA, documents that were irrelevant to the work in question were removed using components such as the abstract, title, keywords, and conclusions.

After this selection, there were 101 articles left, which will be analysed in more detail in the next stage. In order to present the criteria used to remove irrelevant documents and choose the documents closest to the topic, a table was created showing the criteria chosen.

Exclusion Criteria	
Setting	Out of scope of AI, ethic values or scientific research
Subject	Generic and general documents on the subject, which had no specificity
Access	Files that are not open access or require payment to be purchased
Language	All files that are not in English, Portuguese or Spanish

Table 4 : Exclusion Criteria

PREVIEW

Although the exclusion criteria were those highlighted above, the articles removed at this stage were mainly due to the theme and context of the dissertation or lack of access, which makes the language criterion the least used since few articles were recovered that were not in English, Portuguese or Spanish.

For each query, the results were as follows:

Q1	**Q2**	**Q3**	**Q4**	**Q5**
9	22	25	36	9

Table 5 : Number of results per query

At this stage, there was a significant decline in the number of results per query, mainly in queries 1 and 5, due to a large number of articles that were not relevant to the dissertation or that did not cover all the concepts intended in the query.

For each database, the following numbers of articles were selected:

SCOPUS	WoS	EBSCO	ERIC
31	9	49	12

Table 6 : Number of queries per database

Following the same line of comparison and reasoning, in this case, the database that presented the fewest results was Web of Science, and the one that presented the most results was EBSCO.

Since EBSCO is more closely linked to articles that lean toward the more academic and research side, it managed to have a closer relationship with the subject and theme of this dissertation.

2.2.3 PRISMA results

In stage three of PRISMA, the remaining articles were read and analysed, and 19 documents were selected. For this selection, it was necessary to obtain a deeper knowledge and understanding in order to capture the most relevant data for the work in question. The excluded articles, despite presenting important content similar to other documents, proved to be inferior to those selected in terms of the complexity of the work.

The table below contains the inclusion criteria to facilitate the selection of the most relevant and complete articles by comparing them.

Reproduced with permission of copyright owner. Further reproduction prohibited without permission.